

SafeSight × LAISA Digital Transformation Proposal

StudEx Group

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PROPOSAL

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PROPOSAL

SafeSight-LAISA — Digital Transformation PRD

Document: Product Requirements Document **Client:** SafeSight x LAISA **Version:** 1.0 — DRAFT **Date:** 2026-05-12 **Owner:** StudEx Group

1. Executive Summary

SafeSight/LAISA operates a dual-discipline medical practice (ophthalmology + aesthetics) with 14 staff, zero automation, zero AI, paper-based records, and no system integration. They lose an estimated R111,250/month in admin waste, no-shows, and missed revenue opportunities.

This PRD defines a 3-phase transformation that takes them from fully manual operations to AI-assisted intelligent practice management within 6 months, delivering measurable ROI from Month 1.

Core outcome: SafeSight/LAISA becomes the most technologically advanced multidisciplinary clinic in South Africa, running on a StudEx AI Operating System that handles booking, reminders, billing, content, patient communication, and business intelligence — while their 14 staff focus entirely on patient care.

2. Problem Statement

Problem	Quantified Impact	Root Cause
No automation	6.5+ admin hours/day wasted	Manual processes for everything
Paper-based records	Data loss risk, R3K/month printing, slow retrieval	No digital patient management
No-shows (~20%)	R45,000/month in lost revenue	No automated reminder system
No system integration	Triple data entry, errors, no unified view	Elixir Live, My Appointment, Sage are siloed
Manual marketing	4+ hours/day creating content	No content engine
No CRM	Zero patient analytics, no lifecycle tracking	Never implemented
No dashboards	Zero visibility into revenue, patient flow, growth	Data locked in siloed systems
Medical aid verification	1+ hour/day manual calls	No automated verification
Staff bottlenecks at reception	Slow patient flow, frustrated callers	No AI-assisted booking

3. Solution Architecture

PHASE 1: DIGITAL FOUNDATION (WEEKS 1–8)

Goal: Stop the bleeding. Digitise records, automate booking and reminders.

#	Deliverable	What It Does	Acceptance Criteria
1.1	Patient Management System	Replaces paper + Google Drive with searchable, secure digital records	All patient records searchable by name, ID, phone, date
1.2	WhatsApp AI Booking Bot	Handles 80% of booking enquiries automatically	Bot responds within 30 seconds, books appointments into system
1.3	Automated Appointment Reminders	SMS + WhatsApp reminders at 24h and 2h before appointment	No-show rate drops from ~20% to below 12% within 4 weeks
1.4	Digital Patient Intake Forms	QR code + tablet replaces paper consultation/consent forms	New patients fill forms digitally, data auto-enters patient record
1.5	System Integration Layer	Connect Elixir Live ↔ Patient Management ↔ Reminders	Data flows between systems without manual entry

PHASE 2: SMART OPERATIONS (WEEKS 5–14)

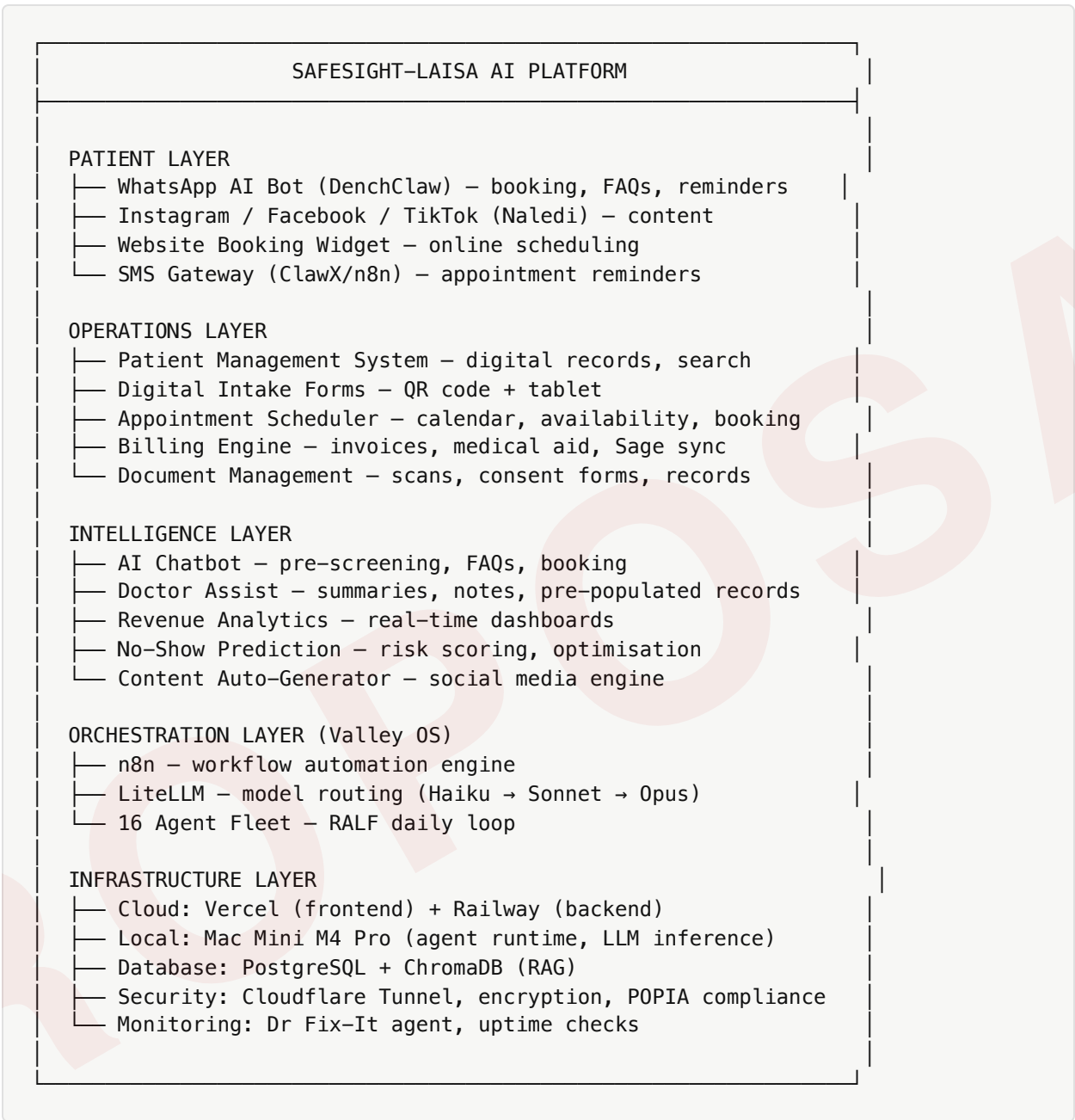
Goal: Automate workflows. Free staff for patient-facing work.

#	Deliverable	What It Does	Acceptance Criteria
2.1	n8n Workflow Engine	Orchestrate all automations across systems	10+ automated workflows running without manual trigger
2.2	Medical Aid Pre-Verification	Check medical aid balance/validity before appointment	80% of medical aid queries resolved automatically
2.3	AI Content Engine	Generate 3 posts/day for Instagram, Facebook, TikTok	Content calendar maintained, posts scheduled automatically
2.4	Automated Billing Flow	Elixir Live → Invoice → Sage Accounting sync	Billing cycle reduced from days to hours
2.5	Post-Appointment Follow-Up	Automated follow-up sequences after visits	Follow-up rate goes from manual/ad-hoc to 100% automated
2.6	New Patient Welcome Sequence	Auto-welcome new patients via WhatsApp	Every new patient receives branded welcome message

PHASE 3: INTELLIGENT PRACTICE (WEEKS 10–24)**Goal:** AI-assisted intelligence. Lead the market.

#	Deliverable	What It Does	Acceptance Criteria
3.1	AI Patient Chatbot	24/7 WhatsApp + website chatbot for FAQs and pre-screening	Handles 70%+ of patient queries without human intervention
3.2	Revenue Analytics Dashboard	Real-time revenue per doctor, per procedure, per day	Dashboard updates hourly, accessible on mobile
3.3	Doctor Support Tools	AI-assisted note-taking, pre-populated records, scan summaries	Doctor consultation prep time reduced by 50%
3.4	Predictive Scheduling	AI predicts no-show risk and optimises calendar	No-show rate drops below 8%
3.5	Treatment Recommendation Engine	Suggests follow-up treatments based on patient history	Increases repeat bookings by 15%

4. Technical Architecture



5. Functional Requirements

FR-001: PATIENT MANAGEMENT

- Store patient demographics, medical history, treatment records

- Search by name, ID number, phone, date of birth
- Link to appointments, billing, and follow-ups
- POPIA-compliant: encryption at rest and in transit, consent tracking

FR-002: WHATSAPP BOOKING

- AI bot handles booking, rescheduling, cancellation
- Bot screens new vs returning patients
- Bot confirms appointment and sends reminder
- Escalates complex queries to reception

FR-003: APPOINTMENT REMINDERS

- 24-hour reminder via WhatsApp
- 2-hour reminder via SMS
- If no response to WhatsApp: Charlie voice call
- No-show flag triggers follow-up sequence

FR-004: DIGITAL INTAKE FORMS

- QR code at reception → patient fills on tablet/phone
- Data auto-enters Patient Management System
- Consent forms digitised and signed electronically
- Eliminates paper entirely

FR-005: BILLING AUTOMATION

- Procedure codes auto-generate invoices
- Medical aid pre-verification before appointment
- Payment tracking and reconciliation
- Sage Accounting sync via n8n

FR-006: CONTENT ENGINE

- 3 posts/day: Instagram, Facebook, TikTok
- Treatment education content
- Before/after showcases (with consent)
- Promotional campaigns
- Performance tracking (engagement, reach, conversions)

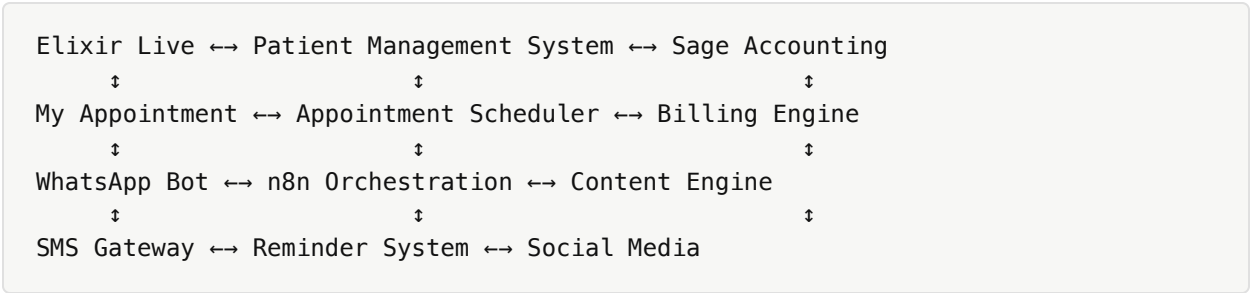
FR-007: ANALYTICS DASHBOARD

- Real-time revenue per doctor, per procedure, per day
- Patient acquisition funnel
- No-show rates and trends
- Marketing ROI
- Monthly/weekly/daily views

6. Non-Functional Requirements

Requirement	Specification
POPIA Compliance	All patient data encrypted at rest (AES-256) and in transit (TLS 1.3). Consent management built-in. Data subject access requests handled within 30 days.
Uptime	99.5% SLA for patient-facing systems (booking, reminders). 99% for internal dashboards.
Performance	Booking bot responds within 30 seconds. Dashboard loads within 3 seconds. Search returns results within 500ms.
Security	POPIA-aligned privacy by design. Role-based access. Audit trails. Regular security reviews by CSO agent.
Backup	Daily database backups. Paper records digitised and stored in 2 locations. Weekly backup verification.
Scalability	Architecture supports adding second location without re-architecture.

7. Integration Map



8. Data Model

PATIENT

- id, name, id_number, phone, email, date_of_birth
- medical_aid_provider, medical_aid_number
- consent_given, consent_date, poopia_status
- created_at, updated_at

APPOINTMENT

- id, patient_id, doctor_id, clinic (eye/aesthetics)
- date, time_slot, status (booked/confirmed/completed/no-show/cancelled)
- reminder_sent_24h, reminder_sent_2h, no_show_flag
- notes, treatment_plan_id

TREATMENT

- id, patient_id, appointment_id, doctor_id
- procedure_code, description, cost
- medical_aid_claimed, payment_status
- follow_up_date, follow_up_status

BILLING

- id, patient_id, appointment_id, treatment_id
- invoice_number, amount, tax
- medical_aid_claim, patient_co_pay
- payment_method, payment_date, reconciliation_status

FOLLOWUP

- id, patient_id, appointment_id
 - type (post_op/routine/treatment_plan)
 - due_date, status, reminder_sent
 - notes, outcome
-

9. Agent Assignment

Agent	Laisa Role	Phase	Primary Model
DenchClaw	Patient Support Bot	1	Claude Haiku
ClawX	Workflow Automation	1-3	Claude Haiku
Robusca	Clinic Coordinator	1	Claude Sonnet
Goose	Documentation & SOPs	1-2	Claude Haiku
CTO	Architecture & Deployment	All	Claude Opus
SkunkWorks	DevOps & Infrastructure	All	Claude Haiku
CSO	Security & POPIA Compliance	All	Claude Sonnet
CashClaw	Revenue & Billing	2	Claude Haiku
Naledi	Content Engine	2	Claude Sonnet
Charlie	Voice Reminders	2	ElevenLabs
CMO	Marketing Strategy	2-3	Claude Sonnet
CAO	Knowledge Management	2-3	Claude Haiku
Hermes	Market Research	3	Claude Sonnet
General	Orchestration	All	Claude Opus
CFO	Budget & ROI Tracking	All	Claude Sonnet
Critic	Quality Assurance	All	Claude Sonnet

10. Success Metrics

Metric	Current	Phase 1 Target	Phase 3 Target
No-show rate	~20%	<12%	<5%
Admin hours/day	6.5h	3h	1h
Booking time per patient	15 min	3 min	1 min
Content posts/week	2-3 (ad hoc)	21 (automated)	21+
Billing cycle time	Days	Hours	Minutes
Revenue visibility	None	Dashboard	Real-time
Patient follow-up rate	Ad hoc	100% automated	100% + predictive
Monthly revenue impact	Baseline	+R18K	+R97.5K

PRD version 1.0 — 2026-05-12 — StudEx Group